AI-Powered Assistive Technologies for the Visually Impaired

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***Abstract*—Assistive AI for blind and visually impaired users has advanced from isolated single-purpose tools narrating individual objects or reading isolated labels toward systems capable of sustaining a full conversational loop grounded in live camera input, spoken language, and personal memory. Yet most deployed solutions still depend on dedicated hardware, mobile applications, or cloud services that treat each interaction as stateless, leaving users without continuity across sessions and without protection against poor-quality visual input that degrades model reliability. This paper presents a voice and vision assistant system: a browser-native, hardware-free real-time assistant designed specifically for blind and visually impaired users, operable on any laptop with a webcam and microphone without application installation or specialised equipment. The system integrates a persistent bidirectional network transport layer, a dual-path vision architecture separating low-latency conversational responses from high-fidelity frame analysis, an eight-class language-model-based intent classifier that routes spoken queries to functionally specialised handlers, a heuristic capture quality coach that enforces frame quality thresholds before vision inference to prevent hallucination on degraded input, and a local semantic memory layer using 1024-dimensional text embeddings with cosine-similarity retrieval for cross-session personalisation. A vision-assisted web search route allows the system to identify physical objects through the camera before generating targeted search queries, extending assistance beyond static scene description into live information retrieval. An online self-correction mechanism logs user feedback signals to adapt response tone and verbosity dynamically across turns without retraining the underlying model. The system is evaluated across seven functional categories spanning scene description, optical character recognition, multilingual translation, object identification, memory recall, web search, and adaptive self-correction, encompassing thirty-eight structured test scenarios that include both successful inference paths and coach-blocked failure cases. Results demonstrate that disciplined turn handling, capture quality gating, and lightweight semantic memory together produce a practically deployable assistive loop that does not require dedicated infrastructure beyond a standard browser session.**

***Index Terms*—assistive technology, blindness, low vision, visual question answering, multimodal systems, retrieval, personalization, capture quality**

I. Introduction

The rapid expansion of multimodal large language models and real-time speech processing capabilities has opened a genuinely new design space for assistive technology — one where a blind or visually impaired user can hold a natural spoken conversation with a system that simultaneously sees, remembers, and searches on their behalf. For the approximately 285 million people worldwide living with visual impairment, daily tasks that sighted individuals perform effortlessly — reading a medicine label, identifying an unfamiliar object, translating a sign in a foreign language, or recalling a fact stored from a previous session — remain disproportionately dependent on either human assistance or narrowly scoped single-function tools that handle one task at a time and retain no memory of the user across interactions.

Early assistive systems addressed individual subtasks in isolation. Screen readers handled text, dedicated OCR applications extracted printed content, and remote sighted-assistance platforms connected blind users with volunteer helpers over video calls. While each approach demonstrated clear value within its intended scope, they share a structural limitation: each requires the user to consciously select the right tool for each situation, switch between applications, and reconstruct context manually across interactions. As generative models matured and multimodal inference became practically viable at

low latency, the research community began exploring whether a single conversational interface could subsume these fragmented capabilities into one coherent assistive loop driven entirely by natural speech.

Recent work on vision-language models and omni-modal architectures has brought that possibility closer to reality, but most published systems still evaluate in controlled laboratory conditions, rely on mobile hardware with dedicated neural processing units, or assume persistent high-bandwidth connections that are unavailable to many users. A further gap that the literature has been slow to address is the problem of degraded visual input. A blind user cannot verify whether the camera is pointed correctly, whether the lighting is adequate, or whether the page is tilted before submitting a frame for analysis. Systems that pass every captured frame to an inference model without quality checking produce hallucinated or meaningless responses that erode user trust, yet no lightweight, speech-integrated quality gating mechanism has been systematically deployed in a browser-native context. Persistent personalised memory across sessions represents a third gap: existing tools treat every interaction as independent, requiring users to re-establish preferences, correct recurring misunderstandings, and repeat contextual facts that the system should have retained.

This paper addresses these three gaps by presenting a voice and vision assistant system built entirely within the browser environment, requiring no dedicated application, no specialised hardware, and no manual tool selection by the user. The proposed system combines real-time spoken interaction, dual-path vision inference, LLM-based intent classification, heuristic frame quality gating, cross-session semantic memory, vision-assisted web search, and adaptive self-correction into a unified assistive loop accessible from any standard laptop.

A. Purpose of This Work

The purpose of this work is to design, implement, and evaluate a browser-native voice and vision assistant that integrates complementary capabilities — perception, speech, memory, and retrieval — into a single conversational interface for blind and visually impaired users, motivated by the fragmentation and deployment limitations identified in existing assistive technology.

B. Motivation

Most existing assistive tools address one capability at a time and retain no state across user sessions, forcing users to manage tool selection and context reconstruction manually. Integrating real-time vision, spoken dialogue, intent routing, capture quality control, and persistent semantic memory into a single browser-native interface removes those coordination burdens and brings practical multimodal assistance within reach of any user with a standard laptop and internet connection.

C. Contributions

The main contributions of this paper are:

- A browser-native, hardware-free voice and vision assistant deployable on any standard laptop without application installation, integrating scene description, OCR, translation, object identification, memory, web search, and self-correction into a single spoken interface.
- A dual-path vision architecture that separates low-latency streaming conversational responses from high-fidelity direct-request frame analysis, routing between paths based on classified user intent.
- An eight-class language-model-based intent classifier that maps free-form spoken queries to functionally specialised handlers without requiring labelled training data from the target deployment domain.
- A heuristic capture quality coach that evaluates incoming frames for luminance sufficiency, local contrast distribution, and spatial uniformity before committing to vision inference, providing spoken corrective guidance when quality thresholds are not met and preventing hallucination on degraded input.
- A local semantic memory layer using 1024-dimensional text embeddings and cosine-similarity retrieval, supporting cross-session fact persistence, priority-fact preloading at session start, and similarity-aware deduplication without requiring external vector database infrastructure.
- A vision-assisted web search route that identifies physical objects through the camera before generating targeted search queries, extending the assistant beyond static description into live information retrieval.
- An online self-correction mechanism that logs user feedback signals and adapts response tone and verbosity dynamically across turns without modifying underlying model weights.

D. Research Questions

This work is guided by the following research questions:

- **RQ1:** Can a voice and vision assistant integrating intent classification, dual-path vision, and semantic memory be deployed effectively in a browser-native environment without specialised hardware?
- **RQ2:** How does heuristic capture quality gating affect the reliability and perceived trustworthiness of vision-based responses under real-world camera conditions?
- **RQ3:** What role does cross-session semantic memory play in reducing the cognitive burden placed on blind users during repeated or preference-sensitive interactions?

II. Literature Review

A. Voice-Only Assistants and Screen Readers

Voice assistants and screen readers provide a foundation for non-visual access. Screen readers are effective for structured digital content, while voice assistants help users perform general tasks through spoken commands. Their main limitation is the absence of camera-grounded understanding. They can answer general questions or read accessible text, but they cannot reliably inspect a medicine label, describe a desk, or determine whether a paper is aligned for OCR without additional visual input.

B. Camera-Based OCR and Scene Description

Camera-based applications extend accessibility into the physical environment. They can read labels, documents, and signs, and some can describe scenes. However, many tools operate as single-purpose applications. The user must choose the task mode, capture the image, and interpret the result. Research on blind-captured images shows that framing, blur, and low light are recurring difficulties, making capture guidance as important as model accuracy [3].

C. Visual Question Answering for Blind Users

VizWiz introduced an interaction model in which blind users capture images and ask questions about them [2]. The later VizWiz dataset formalized the challenge and showed that blind-user images are often unanswerable or lower quality than standard computer-vision datasets [3]. This line of work is central because it shifts evaluation away from generic captioning toward task-specific answers that respond to the user's intent.

D. Remote Sighted Assistance

Remote sighted assistance connects blind users with human helpers through video calls. Studies show that human support can solve tasks that automated systems still struggle with, but they also highlight issues of trust, privacy, timing, and handoff [9]–[11]. These findings matter for automated assistants because they show that guidance style is part of the solution. A system that speaks too much, interrupts poorly, or fails to explain uncertainty may be technically capable but practically frustrating.

E. Personalized Object Recognition and Teachable AI

Generic object recognition is not enough for many assistive tasks. Users often need to recognize personal objects: a specific bottle, a household item, a document, or a familiar package. ReCog and related teachable AI systems show that personalization can improve practical usefulness for blind users [12], [13]. This motivates memory and user-specific context in modern assistants.

F. Vision-Language and Multimodal Models

Vision-language models build on visual question answering, transformers, and cross-modal pretraining [4], [18]–[22]. They can connect text and image features for reasoning tasks. Realtime multimodal APIs extend this further by supporting speech, images, and sometimes video in a conversational loop [27], [30], [31]. For assistive systems, the key design question is not simply which model is strongest, but how models should be routed and guarded in real use.

G. Retrieval, Memory, and Personalization

Retrieval-augmented generation separates stored information from model parameters, making it possible to retrieve facts at inference time [14]. Dense Passage Retrieval and REALM show how retrieval can ground answers in external knowledge [15], [16]. Long-term memory systems such as MemoryBank show how conversational agents can store and reuse user context [17]. For blind-first assistance, memory reduces the need to repeat names, preferences, common objects, and prior corrections.

H. Capture Quality and Interaction Repair

Capture quality is a major difference between benchmark evaluation and real accessibility use. Blind users may not know whether an object is centered or whether a page is readable. Research on blind-captured images and object recognition shows that weak frames reduce reliability [3], [12]. A system should therefore detect poor frames and give spoken repair guidance before attempting detailed inference.

I. Privacy, Uncertainty, and Social Description

Image descriptions can involve people, identity, disability, gender, race, and other sensitive social information. Prior work shows that users want useful details but also need respectful, careful descriptions and uncertainty handling [5], [7]. This is especially important when an assistant describes people in a room or interprets social cues.

III. Comparative Survey Matrix

Table I summarizes major research directions using numerical capability indicators. Values are interpreted as follows: 1 means the capability is central, 0.5 means partial or indirect support, and 0 means mostly absent.

IV. Research Gap Analysis

The review suggests five gaps. First, many systems are functionally fragmented. Second, capture-quality repair is not consistently treated as a first-class component. Third, long-term memory is rarely integrated into blind-first visual assistance. Fourth, realtime conversation is often studied separately from detailed visual reasoning. Fifth, social and identity-sensitive descriptions require careful uncertainty language and user control.

V. Proposed Methodology

A. System Overview

Building on the taxonomy of existing assistive systems for blind users shown in Fig. 1, the proposed system is designed as an integrated, multi-domain voice and vision assistant that addresses the limitations of voice-only assistants, single-task camera applications, navigation-focused aids, and multimodal prototypes that lack frame quality gating or persistent memory. The system follows a multi-stage pipeline consisting of browser-side media capture, network transport, server-side intent classification, dual-path vision routing, frame quality gating, semantic memory retrieval, web search augmentation, spoken response generation, and online self-correction. The detailed component interaction is illustrated in Fig. 2. Each stage is designed to operate entirely within a standard browser session, without requiring dedicated assistive hardware, native application installation, or persistent client-side computing resources.

TABLE I: Literature Comparison Matrix for Assistive Voice and Vision Systems

Research direction	Camera	Speech	Memory	Capture repair	Personalization	Human study	Main contribution
Screen readers and voice assistants	0	1	0.5	0	0.5	1	Access to digital text and general spoken commands.
Camera OCR tools	1	0.5	0	0.5	0	0.5	Physical text reading from documents and labels.
VizWiz and blind-user VQA	1	0.5	0	0.5	0	1	Real user image-question answering with imperfect captures.
Remote sighted assistance	1	1	0.5	1	0.5	1	Human-guided interpretation and repair in live tasks.
Teachable object recognition	1	0.5	1	0.5	1	1	User-specific object recognition and personalization.
Realtime multimodal assistants	1	1	0.5	0.5	0.5	0.5	Low-latency multimodal conversation.
Proposed direction	1	1	1	1	1	0.5	Browser-native integration of routing, repair, memory, and search.

TABLE II: Assistive Technology Category Comparison

Category	Setup burden (1–5)	Visual grounding (0/1)	Stateful memory (0/1)	Repair support (0/1)	Privacy risk (1–5)	Typical limitation
Voice-only assistants	1	0	0.5	0	2	Useful for general queries but not for live physical scenes.
Screen readers	2	0	0	0	1	Strong for accessible digital content, weak for printed objects.
Camera reading apps	2	1	0	0.5	2	Depend on good framing and task-specific modes.
Remote assistance	3	1	0.5	1	4	Human availability and privacy concerns remain.
Multimodal prototypes	2	1	0.5	0.5	3	Often lack deployment-ready memory and repair loops.
Proposed direction	1	1	1	1	2	Requires further user study and threshold calibration.

TABLE III: Research Gap Mapping

Gap	Severity (1–5)	Frequency (1–5)	Proposed priority (%)	Why the gap matters
Fragmented task workflows	5	5	20	Users must switch tools and rebuild context manually.
Poor-frame hallucination	5	4	20	Blind users may act on an answer without seeing the weak input.
Missing persistent memory	4	4	15	Repeated explanations increase speech burden across sessions.
Realtime/detail trade-off	4	4	15	Fast conversation and accurate OCR require different model paths.
Privacy-sensitive description	4	3	15	Descriptions of people and social context can cause harm.
Weak evaluation transfer	3	4	15	Benchmark scores do not always predict assistive usefulness.

B. Browser-Side Media Capture and Transport

User interaction begins at the browser layer, where the system continuously captures microphone audio and camera frames using media access capabilities exposed by modern web browsers. Camera frames are sampled at the instant each new visual frame is rendered, synchronising capture to the browser's display cycle to avoid unnecessary polling and processing overhead. Captured audio and periodic frame snapshots are transmitted to the backend over a persistent, bidirectional network connection, maintaining a full-duplex channel that supports simultaneous audio streaming and image delivery without blocking the spoken interaction loop.

Session lifecycle is coordinated by a server-side session manager that maintains state per user connection for up to 120 minutes, with automatic reconnection negotiated shortly before expiry to prevent mid-task interruption. This transport design keeps the conversational loop stable across extended interactions, without requiring the user to reopen the page or manually re-establish context.

C. Intent Classification

Each spoken turn is first processed by an eight-class language-model-based intent classifier before any routing or inference decision is made. The classifier maps free-form natural language queries to one of eight functional categories: scene description, text reading, translation, object identification, memory saving, memory recall, web search, and general conversation. Classification is performed using a prompt-driven inference call that does not require labelled training data from the target deployment domain, making the system adaptable to new user populations, languages, and tasks without retraining.

The classified intent determines both the processing path and the instruction set supplied to the downstream model, ensuring that scene description turns receive spatially focused guidance while text-reading turns receive layout-aware reading instructions. This decoupling of intent recognition from response

generation allows each functional handler to be developed, tested, and updated independently, without modifying the classification layer or the other handlers.

D. Capture Quality Coach

Before any frame is submitted to a vision inference model, the system applies a lightweight frame quality evaluation that scores three independent signal dimensions: luminance sufficiency, local contrast distribution, and spatial uniformity. Each dimension is compared against a fixed threshold derived from empirical testing across diverse indoor lighting conditions. If any dimension falls below its threshold, the vision call is withheld and the user receives a spoken corrective message specific to the detected deficiency — for example, increasing room lighting, holding the document at a flatter angle, or moving the camera closer to the subject.

This gating mechanism prevents the vision model from producing hallucinated or unreliable responses on frames that are too dark, too blurry, or too spatially uniform to support robust inference [12]. Corrective messages are generated without invoking any large model, keeping the response latency for quality-failed frames below the perceptible threshold and preserving the user's sense of conversational continuity. Only frames that pass all three quality checks are forwarded to the vision inference stage.

E. Dual-Path Vision Architecture

Vision inference is handled through two structurally distinct paths selected based on the classified intent and the conversational context. The streaming path transmits audio and image frames jointly over the persistent bidirectional connection to an omni-modal model optimised for real-time interaction, supporting spoken conversation with embedded visual context at low latency. The deep analysis path issues direct requests to a higher-capacity vision-language model, enabling fine-grained frame analysis for tasks that require precise spatial reasoning, character-level text extraction, or detailed object attribute identification.

Turns classified as scene description or general conversation with low visual complexity are routed to the streaming path, while text reading, translation, object identification, and visually demanding scene description turns are routed to the deep analysis path. This separation ensures that conversational fluency is not sacrificed for turns that do not require exhaustive frame analysis, while turns where accuracy is critical receive the full inference capacity of the more capable model.

F. Semantic Memory Layer

User-specific facts stated during a session are extracted by the memory-saving handler and encoded into 1024-dimensional numerical vector representations using a text embedding model. Each representation is stored alongside its source text, a binary priority flag, session metadata, and a timestamp in a local lightweight relational database, eliminating the need for external vector database infrastructure. Retrieval is performed using cosine-similarity search over all stored representations, returning the top-ranked facts above a fixed similarity threshold.

At the start of each new session, facts marked with the priority flag are retrieved unconditionally and injected into the system instructions before the first user turn, allowing the assistant to greet the user by name or acknowledge known preferences without being explicitly prompted [13]. Similarity-aware deduplication prevents redundant storage of semantically equivalent facts stated across multiple sessions. When a user explicitly corrects a stored fact, the existing entry is updated in place and the revised value takes effect immediately within the same session.

G. Vision-Assisted Web Search

When the classified intent is web search and a camera frame is available, the system first passes the frame through the deep analysis path to identify the subject of the query — for example, the name of a product, the text on a label, or the type of an object — before constructing the search query. This vision-assisted query generation step allows the user to ask questions such as "search the current price of this" while pointing the camera at a product, without needing to name the product explicitly in speech. When no camera frame is relevant to the query, the search route constructs the query directly from the spoken input and retrieves results through a live web search service, which are then summarised and spoken back to the user.

H. Online Self-Correction

The system monitors each spoken turn for feedback signals indicating that a previous response was incorrect, incomplete, or poorly calibrated in length or detail. When a correction signal is detected, a record is written to a dedicated log table in the local database, capturing the session identifier, the intent class of the corrected turn, the signal type, and a timestamp. The session-level correction count and a verbosity state variable are updated accordingly.

When the correction count for a given intent class reaches a threshold within a session, the instruction set for subsequent turns of that intent class is prefixed with a cautionary directive that shifts the model's output toward more hedged and carefully qualified language. Verbosity adjustments — explicit user requests for shorter or longer responses — are stored as session state and applied to all subsequent turns without requiring the user to repeat the instruction. Both correction-driven tone adjustment and verbosity control operate entirely through instruction modification and require no changes to underlying model weights or session infrastructure.

VI. Design Rationale and Architectural Justification

The architectural decisions embedded in the proposed system were each motivated by a specific gap in prior work or a concrete constraint imposed by the target deployment environment. This section justifies the five most consequential design choices.

A. Browser-Native Deployment Without Installation

Prior assistive systems for blind users have predominantly been deployed as native mobile applications, dedicated desktop programs, or hardware-coupled wearables[10]. While these platforms offer low-level hardware access, they impose significant friction on adoption: users must locate, install, and configure software, and updates require active user participation. For blind users, this friction is amplified because the installation process itself may require visual interaction.

The proposed system is accessed entirely through a standard web browser, requiring no installation, no account creation, and no device-specific configuration. The browser exposes microphone and camera access through standardised interfaces available in all major browsers on desktop and mobile devices. This design choice directly addresses the accessibility-of-access problem: if the assistive tool is itself difficult to reach, it fails its primary purpose. The bidirectional full-duplex transport layer ensures that the conversational loop operates at near-native latency despite the browser-mediated architecture.

B. Dual-Path Vision Architecture

A single vision model cannot simultaneously optimise for response latency and inference depth. Streaming omni-modal models deliver responses within hundreds of milliseconds but are trained for conversational fluency rather than character-level precision. High-capacity vision-language models achieve strong performance on fine-grained tasks such as OCR, translation, and detailed object attribute identification, but introduce latency incompatible with real-time spoken interaction.

The dual-path architecture resolves this tension by routing each turn to the path whose cost-accuracy trade-off matches the task. Scene description and general conversation turns are served by the streaming path, preserving the conversational rhythm that blind users depend on. Text reading, translation, and object identification turns are routed to the deep analysis path, where accuracy is the primary requirement and a short additional latency is acceptable. This separation is enforced by the intent classifier at every turn, ensuring that the routing decision is based on the classified task rather than a static per-session configuration.

C. Capture Quality Gating as a First-Class Component

Multimodal assistive systems that rely on blind-captured images face a structural reliability problem: the user cannot verify whether the submitted frame is well-framed, adequately lit, or in focus. Prior work has documented that general-purpose vision models produce hallucinated or confidently wrong responses on low-quality frames, which is particularly harmful in assistive contexts where users may act on the spoken output [12].

The capture quality coach addresses this by evaluating three independent signal dimensions — luminance, local contrast, and spatial uniformity — before any inference call is made. If a frame fails any dimension, the system withholds inference and speaks a corrective instruction to the user. This gating mechanism transforms a silent failure mode into an explicit, actionable feedback loop. Crucially, the corrective response is generated without invoking any large model, keeping the latency for quality-failed frames imperceptibly short.

D. Local Semantic Memory Without External Infrastructure

Conversational assistants that lack persistent memory require users to repeat contextual information at the start of every session — a burden that is disproportionately high for blind users who interact through speech alone. Cloud-based memory solutions introduce privacy concerns and infrastructure dependencies that are inappropriate for a system intended to store personally identifiable health, location, and preference data.

The proposed memory layer stores all user facts locally in a lightweight relational database using 1024-dimensional vector embeddings. No fact leaves the user's machine. Retrieval uses cosine similarity over the stored embeddings, returning contextually relevant facts without requiring exact keyword match. Priority-flagged facts are injected into the system instructions unconditionally at the start of each session, ensuring that the assistant is immediately personalised without requiring the user to re-introduce themselves. This design achieves the personalisation benefits of cloud memory systems while maintaining full local data sovereignty.

E. Online Self-Correction Through Instruction Modification

Model fine-tuning and reinforcement learning from human feedback are well-established methods for adapting language model behaviour, but both require access to model weights and substantial computational infrastructure. For a system deployed at inference time on shared API endpoints, neither approach is available.

The proposed self-correction mechanism operates entirely at the instruction level. When a user signals that a response was incorrect or poorly calibrated, the system logs the correction signal, increments a per-intent correction counter, and modifies the instruction prefix for subsequent turns of that intent class. When the counter reaches a threshold, the instruction shift becomes persistent for the remainder of the session. Verbosity adjustments operate through the same mechanism: a user request for shorter or longer responses updates a session-level verbosity state that is applied to all subsequent system instructions without further prompting. This approach achieves behavioural adaptation without model modification, making it deployable on any inference endpoint regardless of whether the underlying model weights are accessible.

VII. Discussion

The survey shows that assistive usefulness depends on the interaction loop, not only the model. Benchmarks such as MMMU, MathVista, and OCRBench v2 help compare multimodal reasoning, visual math, and text-centric understanding [23]–[25]. However, a blind-first assistant also needs to decide when an answer should not be given. A model may perform well on a benchmark but still fail a user if it confidently describes a dark or misframed image.

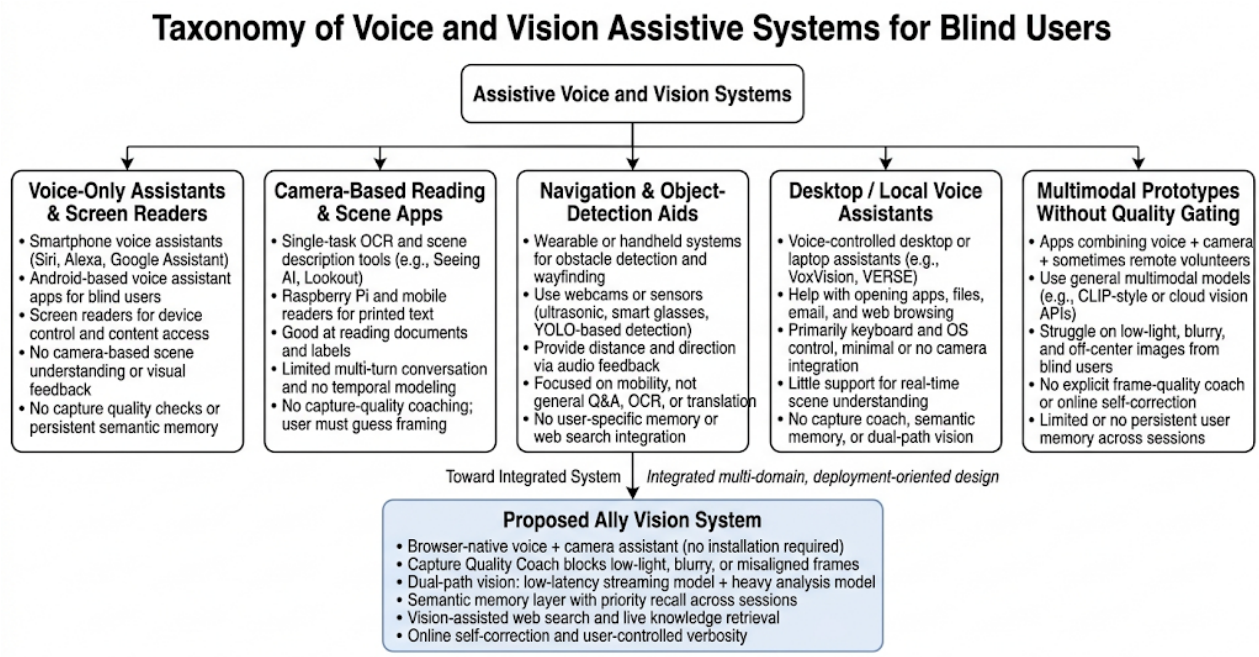

Fig. 1: Taxonomy of voice and vision assistive systems for blind users, positioning the proposed browser-native direction against voice-only assistants, camera reading tools, navigation aids, desktop assistants, and multimodal prototypes.

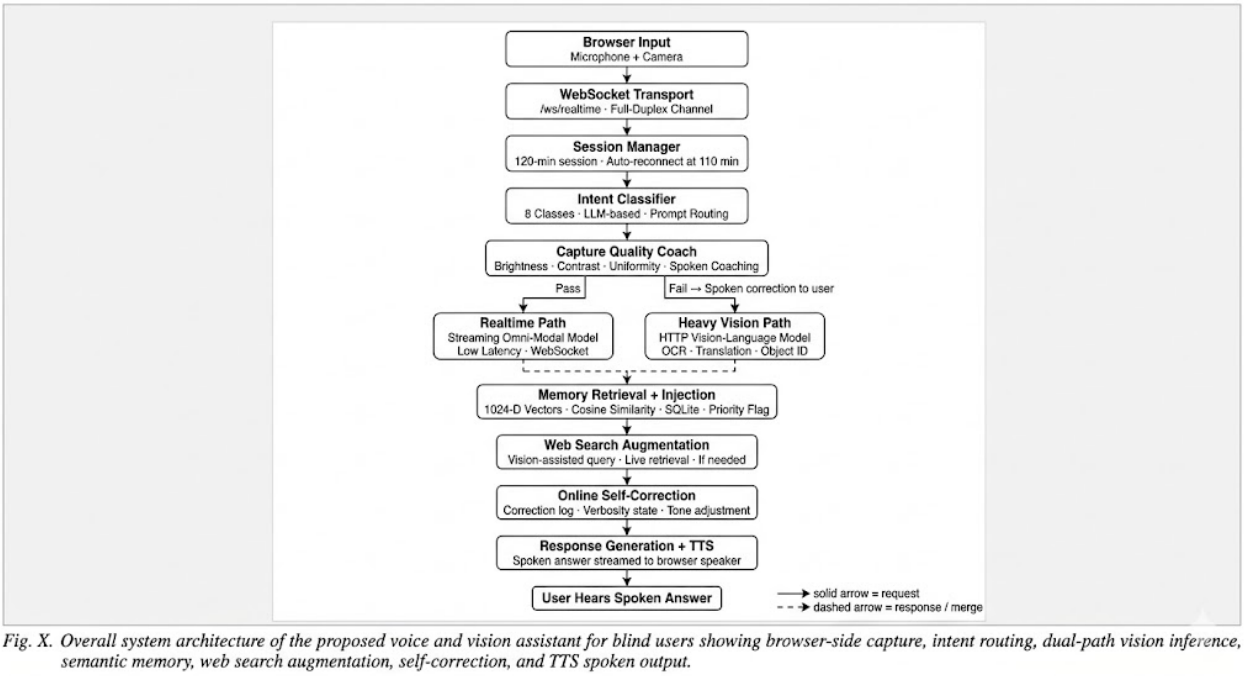

Fig. 2: Architecture direction for AI-Powered Assistive Technologies for the Visually Impaired, showing browser input, WebSocket transport, intent routing, capture-quality coaching, dual-path inference, memory retrieval, web search, self-correction, and spoken response generation.

The strongest direction is therefore not a single model choice but a layered design. Routing separates task types. The capture coach protects vision inference. Memory reduces repeated speech. Search handles current facts. Correction logging gives users a way to shape later responses. This survey positions AI-Powered Assistive Technologies for the Visually Impaired

TABLE IV: Evaluation Lenses for Blind-First Assistance

Task family	Accuracy weight (%)	Latency weight (%)	Human usefulness weight (%)	Risk weight (%)	Key evaluation concern
Scene description and VQA	25	20	35	20	Useful omission control matters as much as object coverage.
OCR and document reading	40	15	25	20	CER/WER must be paired with frame-quality repair.
Remote or guided capture	20	20	30	30	Trust depends on repair instructions and uncertainty handling.
Memory and personalization	30	10	40	20	Saved facts must be accurate, editable, and not overused.
Speech interaction	25	35	25	15	Barge-in, verbosity, and response timing shape usability.

TABLE V: Proposed System Positioning Against Survey Gaps

Design element	Gap coverage (0–1)	Deployment cost (1–5)	User benefit (1–5)	Explanation
Browser-native interface	1	1	5	Reduces installation burden and keeps access simple.
Intent routing	1	2	4	Sends different spoken tasks to specialized handlers.
Capture coach	1	2	5	Prevents answers from poor frames and gives repair guidance.
Dual-path inference	1	3	4	Balances realtime dialogue with accurate visual analysis.
SQLite memory with embeddings	1	2	4	Adds cross-session recall without a hosted vector database.
Self-correction log	0.8	2	4	Stores explicit feedback and adjusts later response behavior.

as a practical architecture that collects these ideas into one coherent prototype direction.

VIII. Challenges and Open Problems

Open problems remain. Capture thresholds must adapt to different cameras and environments. Social description requires policy and user control. Memory systems need editing, deletion, and consent interfaces. Realtime systems must handle barge-in and speech interruption. Benchmark values must be connected to real user outcomes, because technical scores do not directly measure independence, comfort, or trust.

IX. Future Research Directions

Future research should focus on adaptive capture coaching trained on blind-captured frames, multimodal memory that users can inspect and edit, formal user studies with blind and low-vision participants, and evaluation methods that combine benchmark accuracy with human usefulness. Another useful direction is multi-turn task composition, where the user can ask the system to read a label, remember it, and search for related information in a single conversation.

X. Conclusion

This survey reviewed voice and vision assistive technologies for blind and visually impaired users and identified a recurring need for integration. The literature supports the importance of camera-grounded answers, personalized object recognition, remote assistance practices, retrieval, memory, and careful description. At the same time, it shows that fragmented tools, weak capture repair, stateless interaction, and poor uncertainty handling remain problems. AI-Powered Assistive Technologies for the Visually Impaired is presented as a direction that brings these lessons together through a browser-native voice and vision loop with routing, repair, memory, search, and correction.

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